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Hanashima

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(54) **INSTRUMENT PANEL STRUCTURE**

(71) Applicant: **Yazaki Corporation**, Tokyo (JP)

(72) Inventor: **Manabu Hanashima**, Aichi (JP)

(73) Assignee: **YAZAKI CORPORATION**, Tokyo (JP)

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CPC ***B60R 16/0215*** (2013.01); ***B60R 16/0207*** (2013.01); ***B62D 25/145*** (2013.01)

(58) **Field of Classification Search**
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USPC 296/193.02
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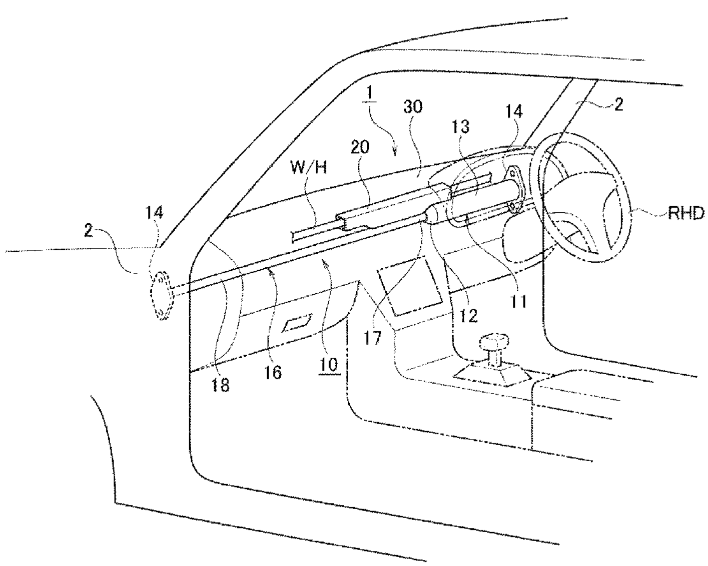
Primary Examiner — Jason S Morrow
(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC

(57) **ABSTRACT**

A reinforcement is bridged between left and right front pillars of a vehicle, and includes: a first pipe arranged at a driver-seat side; and a second pipe having a diameter smaller than a diameter of the first pipe and arranged at a passenger-seat side. A protector is attached to the reinforcement and accommodates a wire harness routed along the reinforcement. An instrument panel is held by the reinforcement and arranged between the left and right front pillars. In a case where a driver seat is positioned on a right side of the vehicle, a locking portion on a right side of the protector is locked in a mounting portion of the first pipe. In a case where a driver seat is positioned on a left side of the vehicle, a locking portion on a left side of the protector is locked in the mounting portion of the first pipe.

3 Claims, 6 Drawing Sheets



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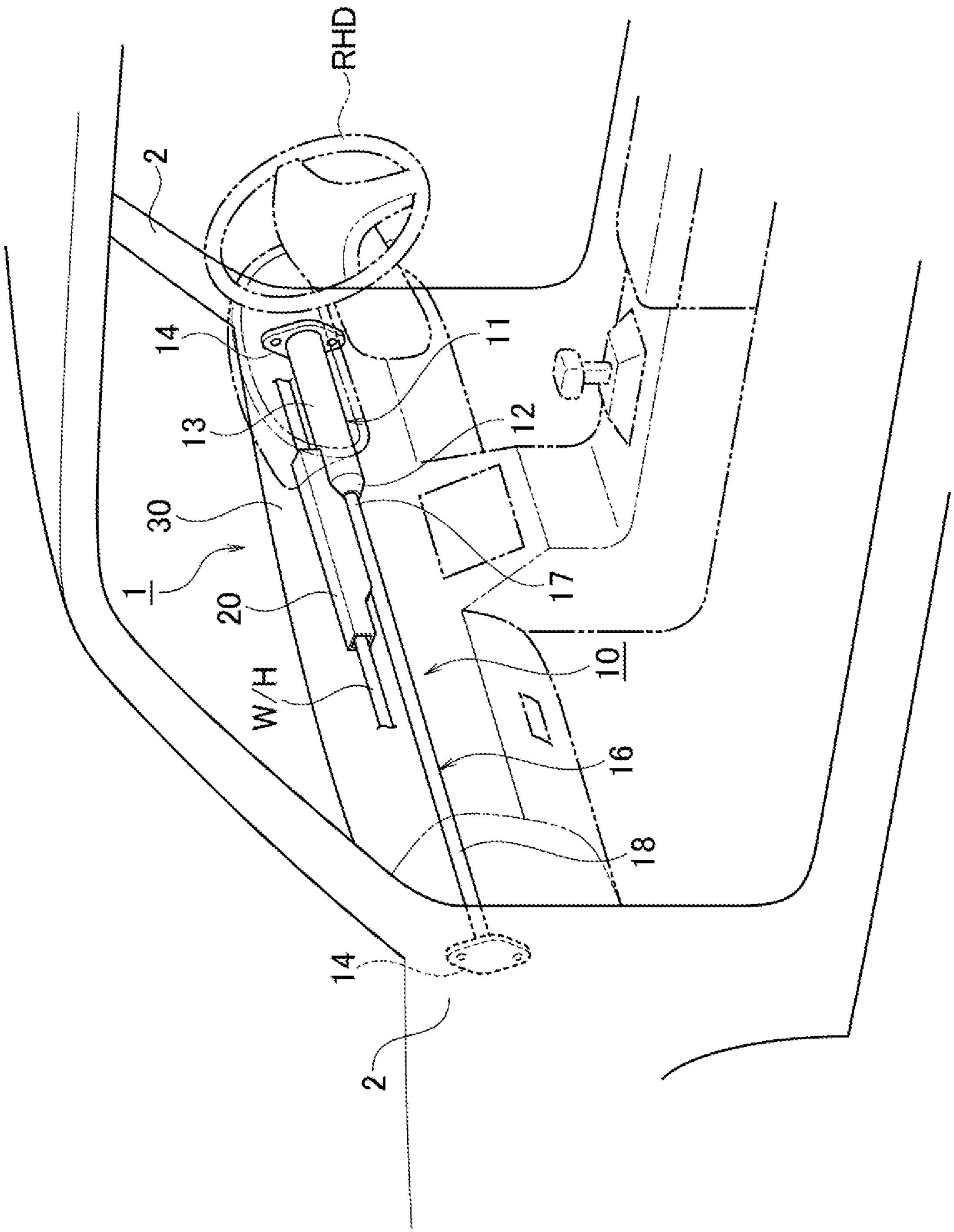
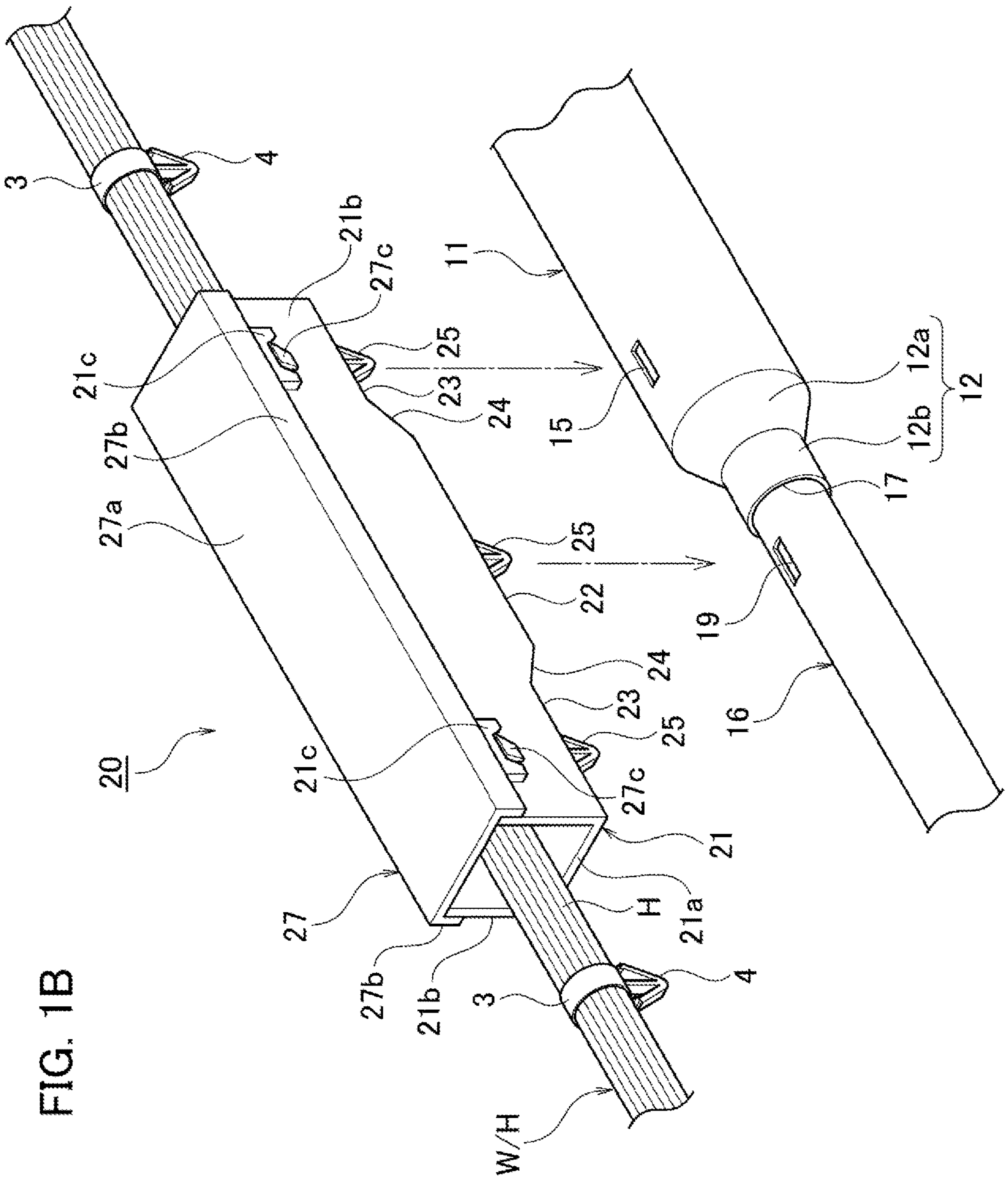


FIG. 1A

FIG. 1B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23
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27b
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21a
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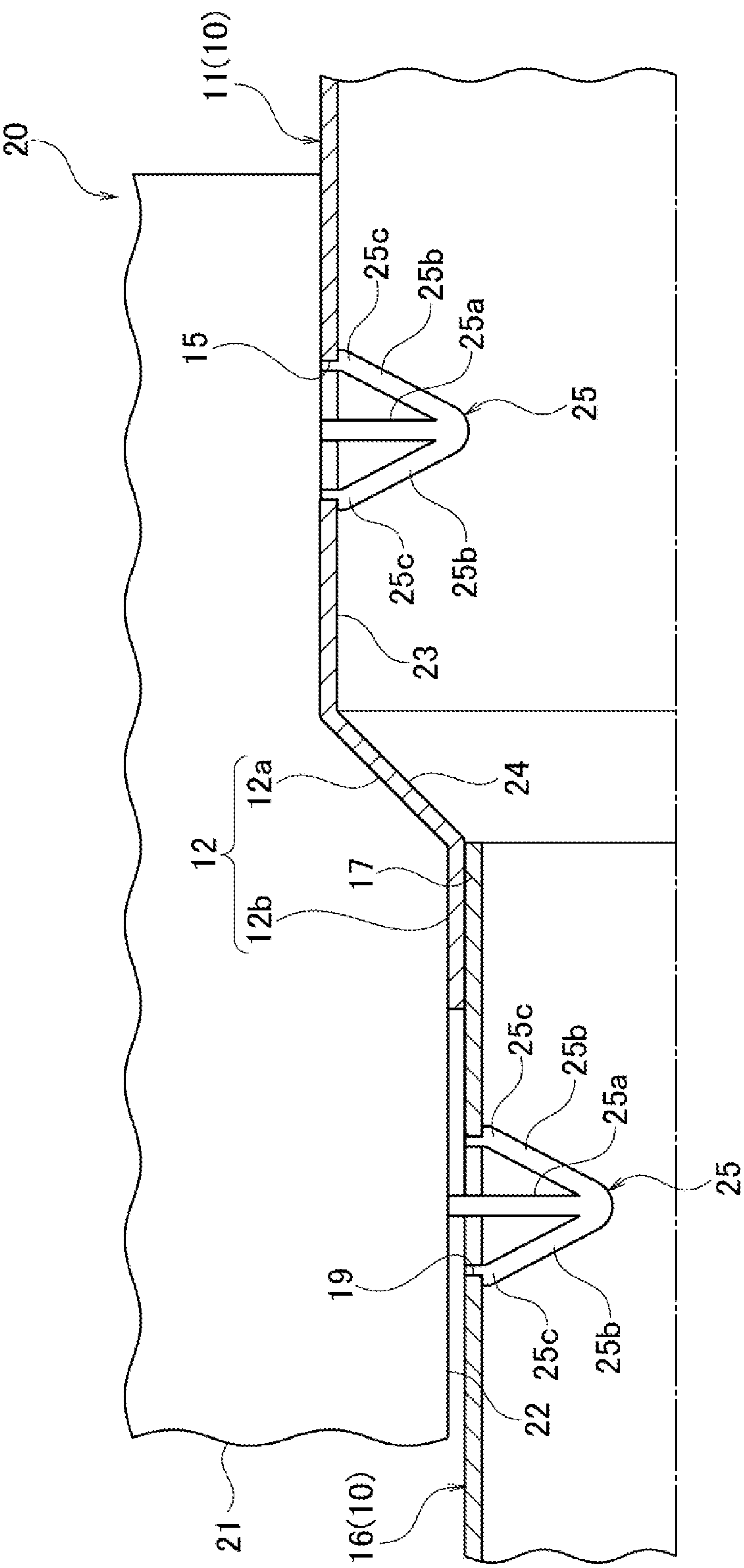
12a
12b
12
15
17
19
16
W/H

FIG. 1C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16(10)

FIG. 2A

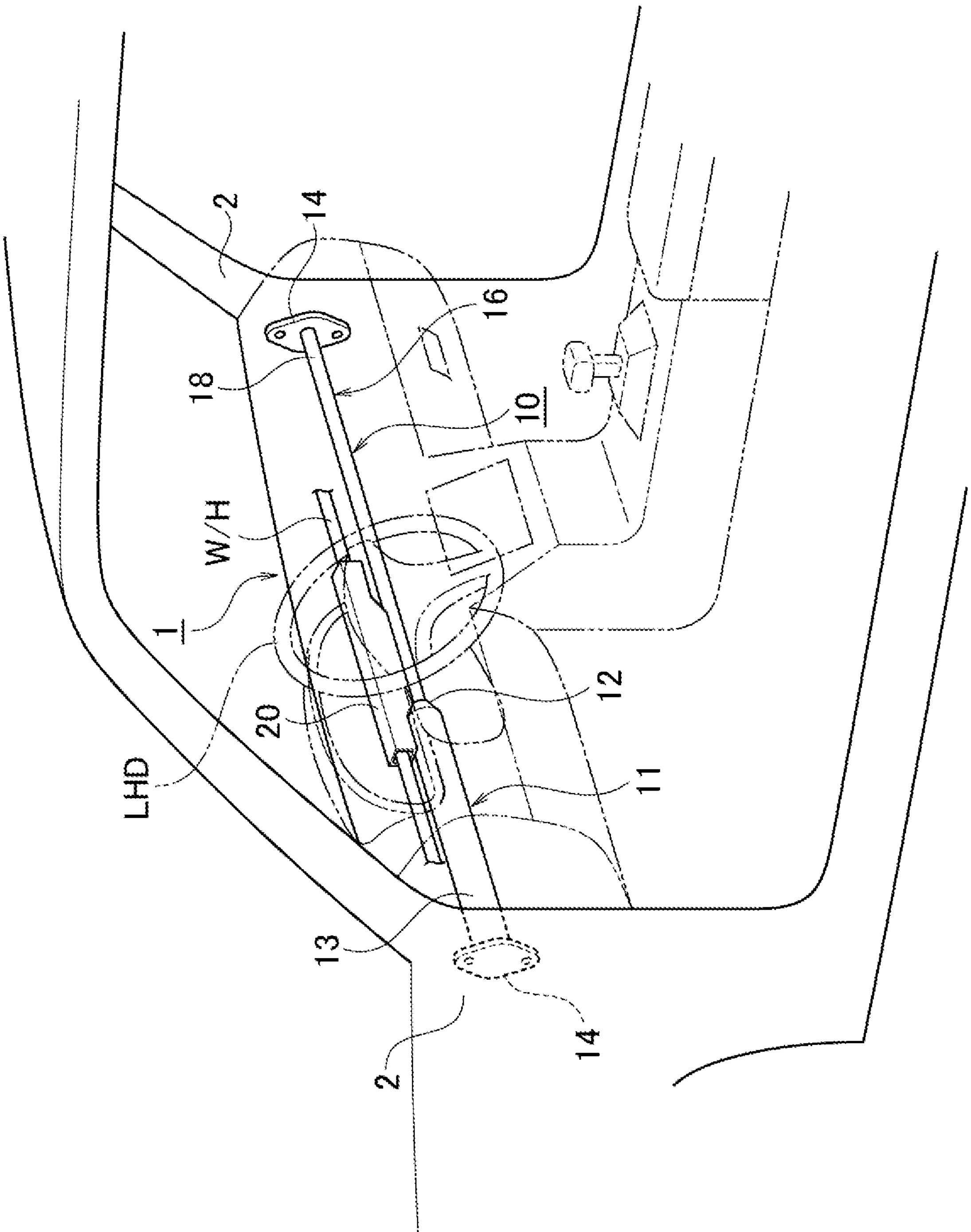
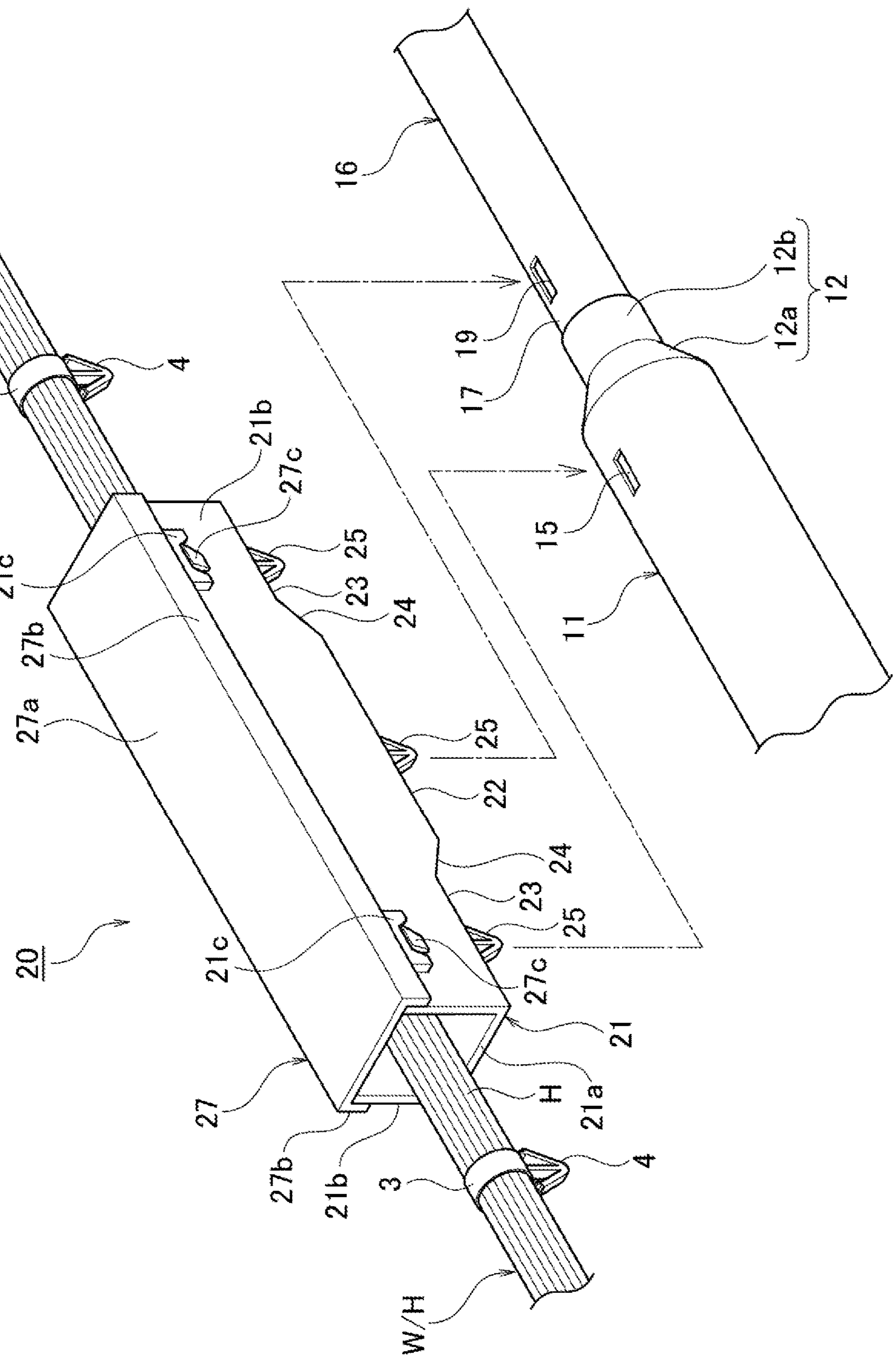


FIG. 2B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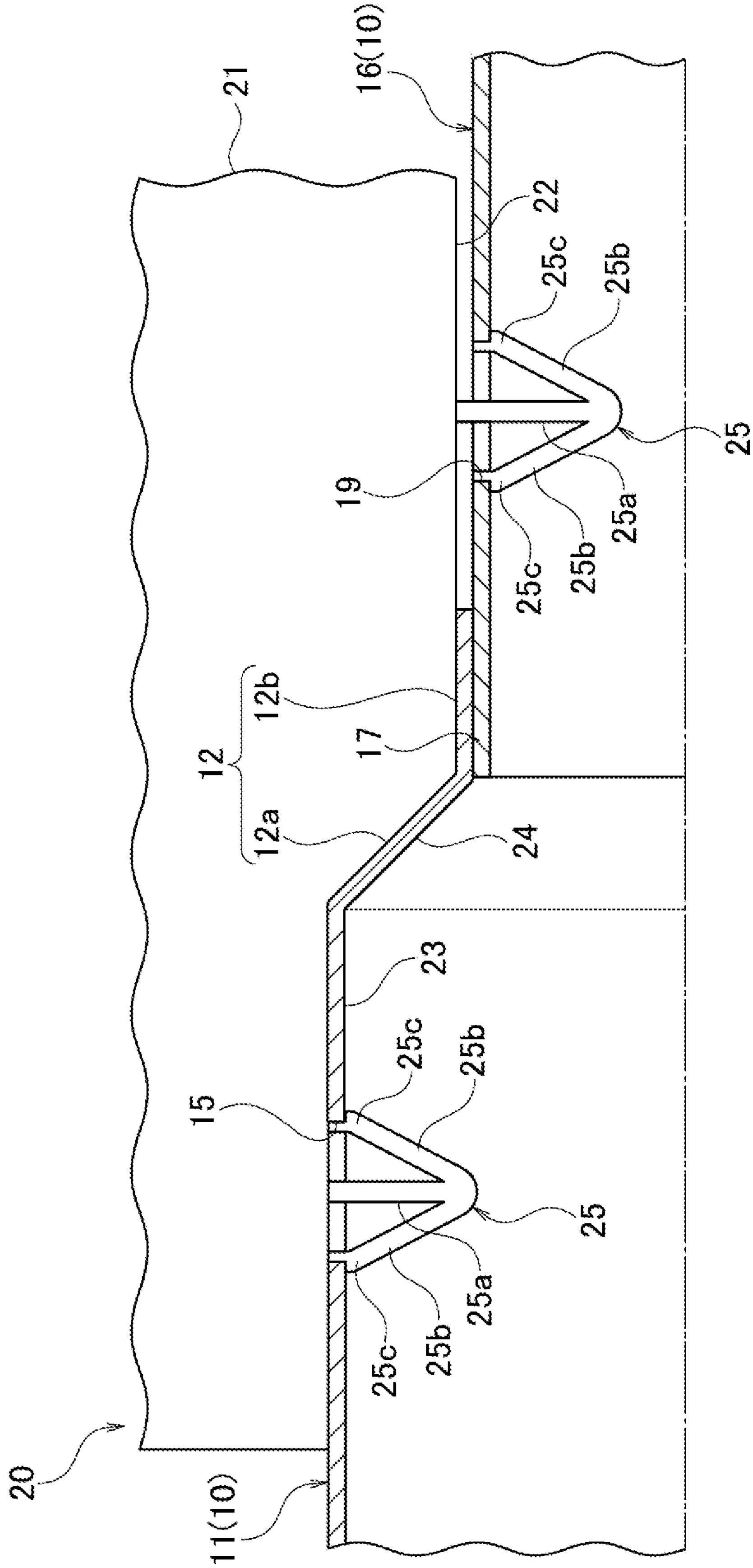
H
W/H

FIG. 2C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25b
25c
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11(10)

INSTRUMENT PANEL STRUCTURE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is based on, and claims priority from the prior Japanese Patent Applications No. 2022-062335, filed on Apr. 4, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The disclosure relates to an instrument panel structure.

BACKGROUND

An instrument panel installed in the interior of an automobile is generally provided with a reinforcement (R/F) as a reinforcing member for the instrument panel. JP 2016-120803 A discloses an instrument panel structure held by a reinforcement. The instrument panel structure is assembled in the reinforcement of a vehicle in which a wire harness is routed, to form the instrument panel portion of the vehicle.

SUMMARY

In the above instrument panel structure, the wire harness is routed along the reinforcement, and thus a protector needs to be provided to protect the wire harness. However, there are two specifications for vehicles: a right-hand-drive type (a type in which the driver seat is arranged on the right side of the vehicle) and a left-hand-drive type (a type in which the driver seat is arranged on the left side of the vehicle). Accordingly, in the course of controlling the wiring route of the wire harness, the development of a protector that can be commonly used for vehicles of two specifications has been desired in view of man-hours and costs.

The disclosure is directed to an instrument panel structure that can be commonly used for both a right-hand-drive type and a left-hand-drive type of vehicles using a protector that is a single component configured to accommodate and protect a wire harness. The instrument panel structure using such a protector makes it possible to achieve low costs by reducing man-hours.

An instrument panel structure in accordance with some embodiments includes: a reinforcement bridged between left and right front pillars of a vehicle, and including a first pipe arranged at a driver-seat side and a second pipe having a diameter smaller than a diameter of the first pipe and arranged at a passenger-seat side; a wire harness routed along the reinforcement; a protector attached to the reinforcement, and accommodating and protecting the wire harness; and an instrument panel held by the reinforcement and arranged between the left and right front pillars. The first pipe includes at least one mounting portion and the second pipe includes at least one mounting portion. The protector includes locking portions at a center and both sides thereof, the locking portions being locked in the mounting portion of the first pipe and in the mounting portion of the second pipe. In a case where a driver seat is positioned on a right side of the vehicle, the locking portion on a right side of the protector is locked in the mounting portion of the first pipe. In a case where a driver seat is positioned on a left side of the vehicle, the locking portion on a left side of the protector is locked in the mounting portion of the first pipe.

According to the above configuration, it is possible to provide an instrument panel structure that can be commonly used for both a right-hand-drive type and a left-hand-drive type of vehicle using a protector that is a single component configured to accommodate and protect a wire harness. The instrument panel structure using such a protector makes it possible to achieve low costs by reducing man-hours.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A is a perspective view illustrating an example of a driver seat of a right-hand-drive vehicle provided with an instrument panel structure for right-hand drive according to an embodiment of the present invention.

FIG. **1**B is a perspective view of the instrument panel structure of FIG. **1**A before a protector is attached to a reinforcement.

FIG. **1**C is a cross-sectional view of the main part of the instrument panel structure in a state in which the protector is attached to the reinforcement of FIG. **1**B.

FIG. **2**A is a perspective view illustrating an example of a driver seat of a left-hand-drive vehicle provided with an instrument panel structure for left-hand drive according to an embodiment of the present invention.

FIG. **2**B is a perspective view of the instrument panel structure of FIG. **2**A before a protector is attached to a reinforcement.

FIG. **2**C is a cross-sectional view of the main part of the instrument panel structure in a state in which the protector is attached to the reinforcement of FIG. **2**B.

DETAILED DESCRIPTION

An instrument panel structure according to an embodiment of the present invention will be described in detail hereinafter with reference to the accompanying drawings.

FIG. **1**A is a perspective view illustrating an example of a driver seat of a right-hand-drive vehicle provided with an instrument panel structure **1** for right-hand drive according to an embodiment of the present invention. FIG. **1**B is a perspective view of the instrument panel structure **1** of FIG. **1**A before a protector **20** is attached to a reinforcement **10**. FIG. **1**C is a cross-sectional view of the main part of the instrument panel structure in a state in which the protector **20** is attached to the reinforcement **10** of FIG. **1**B. FIG. **2**A is a perspective view illustrating an example of a driver seat of a left-hand-drive vehicle provided with an instrument panel structure **1** for left-hand drive according to an embodiment of the present invention. FIG. **2**B is a perspective view of the instrument panel structure **1** of FIG. **2**A before a protector **20** is attached to a reinforcement **10**. FIG. **2**C is a cross-sectional view of the main part of the instrument panel structure in a state in which the protector **20** is attached to the reinforcement **10** of FIG. **2**B.

As illustrated in FIG. **1**A and FIG. **2**A, an instrument panel structure **1** includes: a metal cylindrical reinforcement **10**; a wire harness W/H formed by bundling a plurality of electric wires H; a synthetic resin protector **20**; and a synthetic resin instrument panel **30**.

The reinforcement **10** is formed of a metal cylindrical pipe and is a reinforcing member that increases the rigidity of a vehicle body against external impact. The reinforcement **10** is bridged between left and right front pillars **2** of a vehicle so as to fix a steering wheel and various devices such as an ECU. The reinforcement **10** includes: a cylindrical large-diameter pipe (first pipe) **11** arranged at the driver-seat side; and a cylindrical small-diameter pipe (second pipe) **16** having a diameter smaller than a diameter of the large-diameter pipe **11** and arranged at the passenger-seat side.

The large-diameter pipe **11** and the small-diameter pipe **16** are joined together by means of welding or the like to form the cylindrical reinforcement **10**, which results in a weight reduction and high rigidity of the vehicle. In order to increase the rigidity of the reinforcement **10** at the driver-seat side, the large-diameter pipe **11** is arranged at the driver-seat side. That is, when the rigidity of the driver-seat side is low, the reinforcement **10** deforms and a steering displacement occurs in the event of a vehicle collision; however, the high rigidity of the large-diameter pipe **11** prevents the driver from being in danger in the event of such a vehicle collision.

As illustrated in FIGS. **1**B and **1**C and FIGS. **2**B and **2**C, an inner end **12**, which is one end of the cylindrical large-diameter pipe **11**, is reduced in diameter toward the cylindrical small-diameter pipe **16**. That is, the inner end **12** is configured of: a taper portion **12***a* having an inner diameter that is reduced in diameter in a taper shape to such an extent that the inner diameter is approximately equal to the outer diameter of an inner end **17** which is one end of the small-diameter pipe **16**; and a connecting portion **12***b* which is formed continuously from the taper portion **12***a* and serves as a parallel pipe within a range of a certain length. As illustrated in FIGS. **1**A and **2**A, an outer end **13**, which is the other end of the large-diameter pipe **11**, is provided with a bracket **14** that is bolted to the front pillar **2**. Amounting elongate hole **15** as a mounting portion is provided at a position closer to the inner end **12** between both ends of the large-diameter pipe **11** (between the inner end **12** and the outer end **13**).

As illustrated in FIG. **1**C and FIG. **2**C, the inner end **17**, which is one end of the cylindrical small-diameter pipe **16**, is inserted and welded in an overlapping state in which the inner end **17** comes in contact with the inner circumferential surface of the inner end **12** of the cylindrical large-diameter pipe **11** serving as a parallel pipe. As illustrated in FIG. **1**A and FIG. **2**A, an outer end **18**, which is the other end of the small-diameter pipe **16**, is provided with a bracket **14** that is bolted to the front pillar **2**. A mounting elongate hole **19** as a mounting portion is provided at a position closer to the inner end **17** between both ends of the small-diameter pipe **16** (between the inner end **17** and the outer end **18**).

As illustrated in FIG. **1**B and FIG. **2**B, the wire harness W/H is configured of a bundle of electric wires in which a plurality of electric wires H are bundled together by bands **3** having clamps **4**. The wire harness W/H is inserted into a box-shaped protector **20** extending in the longitudinal direction and routed along the cylindrical reinforcement **10**.

The protector **20** includes a long box-shaped protector body **21** with the upper side and the left and right sides open, and a long box-shaped lid body **27** with the lower side and the left and right sides open. In the protector **20**, the protector body **21** is attached to the reinforcement **10**, and the wire harness W/H is accommodated and protected in the protector **20**.

As illustrated in FIG. **1**B and FIG. **2**B, the protector body **21** of the protector **20** includes a bottom wall **21***a*, and both side walls **21***b* that stand vertically from both sides of the bottom wall **21***a*. A convex portion **22** is provided at the lower surface side at the center of the protector body **21** (that is, at the center of the bottom wall **21***a*). Recesses **23** are provided at the lower surface side on the left and right sides of the protector body **21**, respectively (that is, on the left and right sides of the bottom wall **21***a*). Inclined surfaces **24** that come into contact with the taper portion **12***a* of the large-diameter pipe **11** are provided between the lower surface at the center of the convex portion **22** and the lower surface of each recess **23** on the left and right sides, respectively. Clips **25** as locking portions are provided in a protruding manner in the convex portion **22** at the center and in each recess **23** on the left and right sides, and the clips **25** are locked in the elongate hole **15** of the large-diameter pipe **11** and in the elongate hole **19** of the small-diameter pipe **16**.

As illustrated in FIG. **1**B and FIG. **2**B, a lid body **27** of the protector **20** includes a ceiling wall **27***a*, and both side walls **27***b* that hang from both sides of the ceiling wall **27***a*. The opening of the protector body **21** at the upper surface side is covered with the lid body **27**, and in a state in which locking protrusions **27***c* of the side walls **27***b* of the lid body **27** are locked in lock recesses **21***c* of the side wall **21***b* of the protector body **21**, the wire harness W/H is accommodated and protected therein.

As illustrated in FIG. **1**C and FIG. **2**C, the clips **25** that protrude from the lower surface of the convex portion **22** at the center of the protector body **21** and the lower surfaces of the recesses **23** on the left and right sides of the protector body **21** include a support portion **25***a*, and a pair of elastic pieces **25***b* that are formed in an inverted V-shape from the tip of the support portion **25***a*. In a case where the protector body **21** is attached to the reinforcement **10**, when the pair of inverted V-shaped elastic pieces **25***b* of the clip **25** are respectively inserted into the mounting elongate hole **15** of the large-diameter pipe **11** and the mounting elongate hole **19** of the small-diameter pipe **16**, the pair of elastic pieces **25***b* elastically deform toward the support portion **25***a* side. Thereafter, when each of L-shaped locking protrusions **25***c* provided at the tip of the pair of elastic pieces **25***b* is locked to the short edges of the mounting elongate holes **15** and **19**, this locking state is maintained by the elastic biasing force of the pair of elastic pieces **25***b*, and the clips **25** do not come out of the mounting elongate holes **15** and **19**.

As illustrated in FIG. **1**B, in the case of right-hand drive RHD (in the case where the driver seat is positioned on the right side of a vehicle), the clip **25** that is provided in a protruding manner in the right-side recess **23** of the protector **20** is locked in the mounting elongate hole **15** of the large-diameter pipe **11** arranged at the driver-seat side of the reinforcement **10**. As illustrated in FIG. **2**B, in the case of left-hand drive LHD (in the case where the driver seat is positioned on the left side of a vehicle), the clip **25** that is provided in a protruding manner in the left-side recess **23** of the protector **20** is locked in the mounting elongate hole **15** of the large-diameter pipe **11** arranged at the driver-seat side of the reinforcement **10**.

The instrument panel **30** is made of synthetic resin. The instrument panel **30** is held by the reinforcement **10** and arranged between the left and right front pillars **2** in such a way as to cover the protector **20** and the wire harness W/H.

As described above, according to the instrument panel structure **1** of the present embodiment, as illustrated in FIG. **1**B, in the case of right-hand drive RHD, the clip **25** that is provided in a protruding manner in the right-side recess **23** of the protector **20** is locked in the mounting elongate hole **15** of the large-diameter pipe **11** arranged at the driver-seat side of the reinforcement **10**. Further, the clip **25** that is provided in a protruding manner on the convex portion **22** at the center of the protector **20** is locked in the mounting elongate hole **19** of the small-diameter pipe **16** arranged at the passenger-seat side of the reinforcement **10**. Since these clips **25** are locked in this way, the protector **20** accommodating the wire harness W/H is attached onto the reinforcement **10**. At this time, as illustrated in FIG. **1**C, the inclined surface **24** between the convex portion **22** and the right-side recess **23** of the protector **20** is in contact with the tapered portion **12***a* of the large-diameter pipe **11** of the reinforcement **10** without a gap, and the bottom surface of the convex portion **22** of the protector **20** is in contact with the connecting portion **12***b* of the large-diameter pipe **11** without a gap. These contacts make it possible to hold the wire harness W/H at both ends of the protector **20**. Although only the two clips **25** out of the three clip **25** of the protector **20** are fixed to the reinforcement **10**, the bundle of thick electric wires of the wire harness W/H passing through the protector **20** ensures the rigidity of the protector **20**, thereby preventing interference with peripheral components.

As illustrated in FIG. **2**B, in the case of left-hand drive LHD, the clip **25** that that is provided in a protruding manner in the left-side recess **23** of the protector **20** is locked in the mounting elongate hole **15** of the large-diameter pipe **11** arranged at the driver-seat side of the reinforcement **10**. Further, the clip **25** that is provided in a protruding manner on the convex portion **22** at the center of the protector **20** is locked in the mounting elongate hole **19** of the small-diameter pipe **16** arranged at the passenger-seat side of the reinforcement **10**. Since these clips **25** are locked in this way, the protector **20** accommodating the wire harness W/H is attached onto the reinforcement **10**. At this time, as illustrated in FIG. **2**C, the inclined surface **24** between the convex portion **22** and the left-side recess **23** of the protector **20** is in contact with the tapered portion **12***a* of the large-diameter pipe **11** of the reinforcement **10** without a gap, and the bottom surface of the convex portion **22** of the protector **20** is in contact with the connecting portion **12***b* of the large-diameter pipe **11** without a gap. These contacts make it possible to hold the wire harness W/H at both ends of the protector **20**. Although only the two clips **25** out of the three clip **25** of the protector **20** are fixed to the reinforcement **10**, the bundle of thick electric wires of the wire harness W/H passing through the protector **20** ensures the rigidity of the protector **20**, thereby preventing interference with peripheral components.

As described above, the present disclosure makes it possible for a protector **20** that is a single component configured to accommodate and protect a wire harness to be commonly used for both a right-hand-drive RHD type and a left-hand-drive LHD type of vehicles. Therefore, it is possible to achieve low costs by reducing man-hours because of such a protector.

In the above embodiment, the large-diameter pipe **11** and the small-diameter pipe **16** are joined together to form the cylindrical reinforcement **10**; however, the present embodiment is not limited to this configuration. For example, it may be possible to form the cylindrical reinforcement **10** by integrally molding a flat plate having a wide section that is a large-diameter section and a narrow section that is a small-diameter section by means of bending processing or the like.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

The invention claimed is:

1. An instrument panel structure comprising:

a reinforcement bridged between left and right front pillars of a vehicle, and including a first pipe arranged at a driver-seat side and a second pipe having a diameter smaller than a diameter of the first pipe and arranged at a passenger-seat side;

a wire harness routed along the reinforcement;

a protector attached to the reinforcement, and accommodating and protecting the wire harness; and

an instrument panel held by the reinforcement and arranged between the left and right front pillars, wherein

the first pipe includes at least one mounting portion and the second pipe includes at least one mounting portion,

the protector includes locking portions at a center and both sides thereof, the locking portions being locked in the mounting portion of the first pipe and in the mounting portion of the second pipe,

in a case where a driver seat is positioned on a right side of the vehicle, the locking portion on a right side of the protector is locked in the mounting portion of the first pipe, and

in a case where a driver seat is positioned on a left side of the vehicle, the locking portion on a left side of the protector is locked in the mounting portion of the first pipe.

2. The instrument panel structure according to claim **1**, wherein

the first pipe includes a mounting elongate hole as the mounting portion and the second pipe includes a mounting elongate hole as the mounting portion,

the protector includes a convex portion positioned at a lower surface side at the center of the protector and recesses positioned at the lower surface side on left and right sides of the protector, and

the protector includes clips as the locking portions, the clips protruding from each of the recesses on the left and right sides and from the convex portion and locked in the mounting elongate hole of the first pipe and in the mounting elongate hole of the second pipe.

3. The instrument panel structure according to claim **2**, wherein

the reinforcement includes a taper portion reduced in diameter in a taper shape between the first pipe and the second pipe, and

the protector includes inclined surfaces between a lower surface of the convex portion and a lower surface of each of the recesses on the left and right sides, the inclined surfaces coming into contact with the taper portion.

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